

Kushagra Jaiswal

ML Engineer & Research Lead

Faridabad, Haryana, India | kinuj768@gmail.com | [linkedin.com/in/kushagra356](https://www.linkedin.com/in/kushagra356) | github.com/Nerd-coderZero | nerd-coderzero.github.io

Open to remote, hybrid, or relocation

SUMMARY

ML Engineer and Research Lead with experience designing and delivering machine learning systems across healthcare AI, NLP, computer vision, and time series forecasting. Translates research into working implementations — including CNN-Transformer hybrids, CBAM attention, federated learning pipelines, and retrieval-augmented generation. Led a small research team while personally shipping 20+ projects end-to-end.

EXPERIENCE

ML Engineer & Research Lead

2024 — Present

Megaminds IT Services — Remote

- Architected and delivered 20+ ML systems across medical imaging, NLP, industrial time series, and LLM integration.
- Implemented research architectures into working systems: CNN-Transformer hybrids with cross-attention fusion, CBAM spatial/channel attention on ResNet backbones, bidirectional LSTM for anomaly detection, and federated learning pipelines.
- Achieved 97% accuracy on histopathology image classification, 89% on multi-class MRI classification, and 86.5% macro F1 on irony/sarcasm detection with severe class imbalance.
- Led a team of 10+ research associates: project assignment, technical mentorship, onboarding, and performance review.
- Managed client communication and requirements across delivered projects.
- Applied AI-assisted development workflows to reduce implementation time on standard tasks.
- Domains covered: medical imaging, predictive maintenance (SCADA), NLP classification, RAG pipelines, GAN-based augmentation, AutoML, federated learning.

SELECTED PROJECTS

Breast Cancer Detection — 97% Accuracy

End-to-end histopathology image classification pipeline. Combined EfficientNet-B3 convolutional feature extraction with Transformer cross-attention for a hybrid architecture capturing local texture and global context. Outperformed CNN-only baseline by +11 percentage points on held-out test data.

Stack: PyTorch, timm, MONAI | Dataset: BreakHis (7,909 images, 4 magnifications)

Brain Tumor MRI Classification — 89% Accuracy

4-class MRI classification system using a ResNet backbone augmented with Convolutional Block Attention Module (CBAM) for spatial and channel-wise feature selection on diagnostically relevant regions.

Stack: TensorFlow, Keras

Twitter Irony & Sarcasm Detection — 86.5% Macro F1

Multi-modal NLP classifier handling severe label imbalance (1:9 ratio). Fine-tuned BERT on tweet text, fused with engagement metadata. Used focal loss and post-hoc threshold calibration; improved macro F1 from 71% to 86.5%.

Stack: HuggingFace Transformers, PyTorch

Wind Turbine Predictive Maintenance

Multivariate time series model on SCADA sensor data to detect degradation patterns before failure. Bidirectional LSTM with temporal attention weights.

Stack: PyTorch, Scikit-learn, Pandas

Domain-Adaptive RAG Chatbot *(Open Source)*

Configurable RAG pipeline for domain-specific Q&A. Document ingestion, FAISS vector indexing, retrieval strategy comparison, and Gemini-powered generation. Includes evaluation harness.

Stack: LangChain, FAISS, Gemini API, Python

<https://github.com/Nerd-coderZero/rag-chatbot>

IPL Match Outcome Prediction

Pre-match win prediction for IPL cricket using team-level historical features: phase-wise run rates, win-rate differentials, death-over economy, and venue head-to-head stats. Three classifiers evaluated with strict temporal split (2008-2018 train, 2019-2024 test): Logistic Regression AUC 0.617, Random Forest 0.604, XGBoost 0.600. SHAP explainability and ablation study confirmed phase-wise run rates as the most predictive feature group.

Stack: Python, XGBoost, Scikit-learn, SHAP, Optuna, pandas

eBPF ML Security Inference Engine

Real-time intrusion detection system using eBPF kernel event telemetry, deployed as a containerised ML inference service on Kubernetes (Kind). Ensemble of Random Forest, LSTM Autoencoder, and Isolation Forest detects data exfiltration, lateral movement, and cryptomining patterns in synthetic test scenarios. Models baked into Docker image; exposed as REST API with sub-second inference latency.

Stack: Python, Kubernetes, Docker, Random Forest, LSTM, Isolation Forest, REST API, eBPF

TECHNICAL SKILLS

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, timm, MONAI, LangChain

Architectures: CNN-Transformer hybrids, Vision Transformers (ViT), CBAM attention, Bidirectional LSTM, LSTM Autoencoder, BERT fine-tuning, RAG pipelines, Isolation Forest, Federated Learning, GANs, AutoML

Domains: Medical Imaging, NLP & LLMs, Time Series Forecasting, Anomaly Detection, Retrieval-Augmented Generation, Intrusion Detection, Sports Analytics

Evaluation & Experimentation: ROC/AUC, Macro/Micro F1, SHAP explainability, Ablation studies, Optuna hyperparameter tuning, Threshold calibration, Temporal train/test splits

Tools & Infrastructure: Python, Git, Docker, Kubernetes, REST API, FAISS, Gemini API, Jupyter, GitHub Pages

EDUCATION

BTech, Computer Science Engineering J.C. Bose University of Science and Technology, YMCA	2017 – 2021
Post Graduate Program — Data Science UpGrad	2023

KEY METRICS

- 20+ ML projects delivered end-to-end
- 10+ research associates onboarded and managed
- 97% accuracy on medical imaging (held-out test set)
- 86.5% macro F1 on heavily imbalanced NLP task